Designing an End-to-End Machine Learning Pipeline

Customer Churn Prediction and Retention Strategy

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1. Executive Summary

Customer churn represents one of the most significant financial risks for telecommunications companies operating in competitive markets. Acquiring a new customer costs five to seven times more than retaining an existing one, making accurate churn prediction a high-value business capability. This report presents the design, implementation, and evaluation of an end-to-end machine learning pipeline for predicting customer churn using the BigML telecom dataset.

The project follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework and evaluates four distinct models: Logistic Regression, Random Forest, XGBoost, and a deep learning Neural Network (Multi-Layer Perceptron). Each model was trained on the same preprocessed feature set comprising 23 engineered variables derived from 20 raw customer attributes covering usage, billing, and engagement patterns.

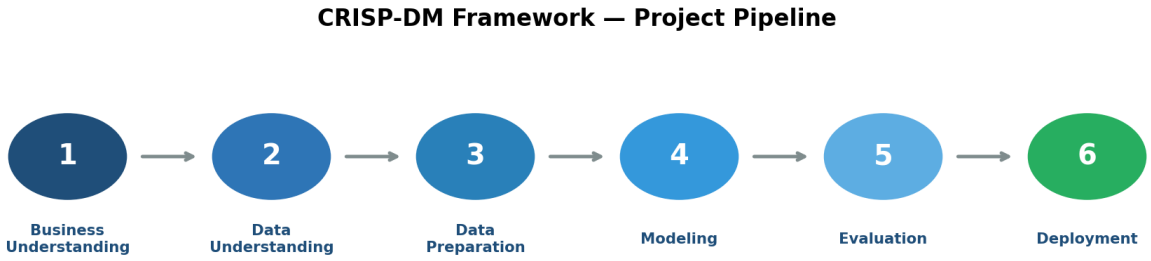

Figure 1: CRISP-DM Framework guiding the project pipeline.

XGBoost emerged as the top-performing model with 97.6% accuracy, 100% precision, 84% recall, and an F1 score of 0.91 for the churn class. It correctly identified 66 of 79 actual churners with zero false positives, meaning every customer flagged for retention intervention was a genuine churn risk. The key drivers of churn identified across all models were: international plan subscription, customer service call frequency (≥3 calls), and total daytime usage and charges.

Based on these findings, we recommend deploying XGBoost as the primary churn scoring model, supplemented by Logistic Regression coefficients for interpretability. With an estimated customer lifetime value of $1,200–$2,000 and retention campaign costs of $50–$100 per intervention, this pipeline can generate a projected **net annual benefit of $3.2M–$6.1M** for a mid-size telecom operator through targeted, data-driven retention efforts.

2. Problem Statement and Research Objectives

2.1 Problem Statement

Customer churn represents a significant financial risk for organizations in competitive markets. While companies collect extensive data across billing, usage, and engagement channels, these data are often underutilized or analyzed in isolation. The core challenges include data silos where customer data across channels is analyzed separately, the absence of timely early-warning mechanisms to identify at-risk customers, and the lack of scalable, automated analytical pipelines for continuous churn monitoring.

2.2 Research Objectives

- Design an end-to-end ML pipeline for churn prediction using the CRISP-DM framework.
- Explore and engineer customer-level features capturing behavioral, temporal, and transactional patterns.
- Develop and evaluate multiple ML and deep learning models using appropriate performance metrics.
- Identify the most influential drivers of customer churn and assess their business implications.
- Propose a feasible deployment and automation strategy for ongoing churn monitoring.

2.3 Research Framework and Key Assumptions

The unit of analysis is the individual customer, where each row represents one customer's complete behavioral profile. The outcome variable is a binary churn indicator (Churn vs. No Churn). The methodology follows the CRISP-DM Framework with six iterative phases.

Key Assumptions: Historical customer behavior is predictive of future churn; churn patterns are stable enough to support supervised learning; and retention interventions can be prioritized based on predicted risk scores.

3. Literature Review

This review synthesizes nine key sources across three thematic areas: methodological foundations, churn prediction and profitability, and advanced techniques and applications.

3.1 Methodological Foundations

Shearer (2000) introduced the CRISP-DM framework, providing a six-phase iterative methodology encompassing business understanding, data understanding, data preparation, modeling, evaluation, and deployment. This framework emphasizes aligning technical analysis with business objectives and acknowledges the inherently iterative nature of machine learning projects. Our pipeline directly follows this methodology.

Domingos (2012) argued that feature quality matters more than algorithm choice and discussed the bias-variance tradeoff, overfitting risks, and the no-free-lunch theorem. This insight justifies our multi-model evaluation approach rather than reliance on any single algorithm.

Sculley et al. (2015) highlighted hidden technical debt in ML systems, including poorly documented pipelines, brittle feature dependencies, data drift, and lack of monitoring. Their work informed our emphasis on designing robust, maintainable pipelines with version control and reproducibility through fixed random states.

3.2 Churn Prediction and Profitability

Verbeke et al. (2012) introduced profit-driven evaluation for churn models, demonstrating that traditional metrics such as accuracy and AUC are insufficient because they ignore the asymmetric costs of churn misclassification. Models with similar accuracy can produce very different business outcomes when measured by expected profit from targeted retention campaigns.

Neslin et al. (2006) evaluated churn models by their ranking ability rather than raw accuracy, arguing that ordering customers by defection risk is more aligned with how managers deploy retention resources. Even modest improvements in predictive ranking can yield substantial financial gains at scale.

Burez and Van den Poel (2009) systematically evaluated resampling, cost-sensitive learning, and ensemble methods for imbalanced churn datasets. Their work supports our use of recall, precision, F1-based evaluation, and class-weight adjustments to address the 14.5% minority churn class.

3.3 Advanced Techniques and Applications

Lemmens and Croux (2006) demonstrated that ensemble methods, particularly bagging and boosting, significantly outperform single classifiers for churn prediction. Mozer et al. (2000) pioneered the use of neural networks for telecom churn, demonstrating that raw data must be aggregated into meaningful behavioral features for effective deep learning. Imani et al. (2025)

provided a systematic review of over 200 studies, synthesizing trends in ensemble, deep learning, and hybrid methods. Ullah et al. (2019) demonstrated a hybrid model combining classification and clustering that achieved approximately 88.6% accuracy for telecom churn.

3.4 Key Themes from the Literature

- **Feature engineering over algorithm choice:** Multiple studies confirm that feature quality and business relevance have greater impact than algorithm selection alone.
- **Profit-driven evaluation:** Traditional accuracy metrics are insufficient; models must be evaluated by ranking ability and expected profit impact.
- **Class imbalance is critical:** Proper handling via resampling, cost-sensitive learning, and appropriate metrics is essential.
- **Business interpretability matters:** Model outputs must be translated into actionable retention strategies for executive stakeholders.

4. Exploratory Data Analysis

4.1 Dataset Overview

The analysis uses the BigML telecom churn dataset containing 2,666 customer records with 20 features. The dataset includes a mix of numeric features (account length, usage minutes, charges, call counts) and categorical features (state, area code, international plan, voice mail plan). The target variable is Churn (boolean). No missing values were found across any column.

Feature	Type	Description
Account length	int64	Duration of customer relationship
International plan	binary	Subscribed to international plan (Yes/No)
Voice mail plan	binary	Has voicemail plan (Yes/No)
Total day/eve/night minutes	float64	Usage minutes across time periods
Total day/eve/night charge	float64	Charges per time period
Customer service calls	int64	Number of support calls made
Churn (target)	boolean	Whether the customer churned

Table 1: Key features in the telecom churn dataset.

4.2 Target Distribution and Class Imbalance

The churn distribution reveals a significant class imbalance: 2,278 customers (85.5%) did not churn while only 388 customers (14.5%) churned, representing an approximately 6:1 imbalance ratio. A naive model predicting No Churn for every customer would achieve 85.5% accuracy while catching zero actual churners. This guided our decision to use class weights, SMOTE, and to prioritize recall and F1 score as evaluation metrics.

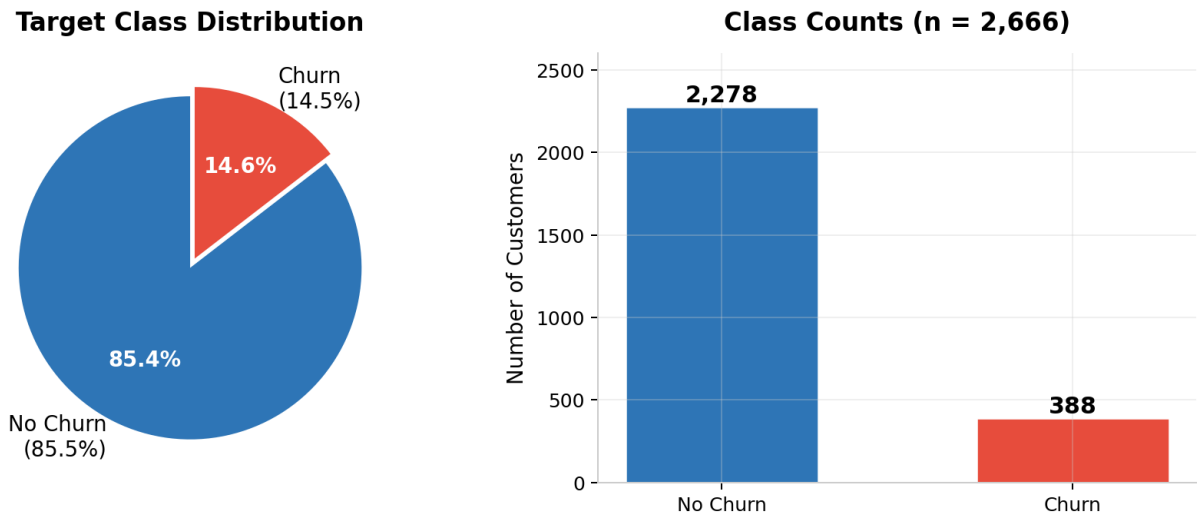

Figure 2: Target class distribution showing 85.5%/14.5% imbalance.

4.3 Key Correlations and Churn Predictors

Correlation analysis identified International plan (+0.285), high service calls (+0.252), Customer service calls (+0.245), and Total day minutes (+0.206) as the top positive correlators with churn. Voice mail plan (–0.166) emerged as a protective factor associated with lower churn.

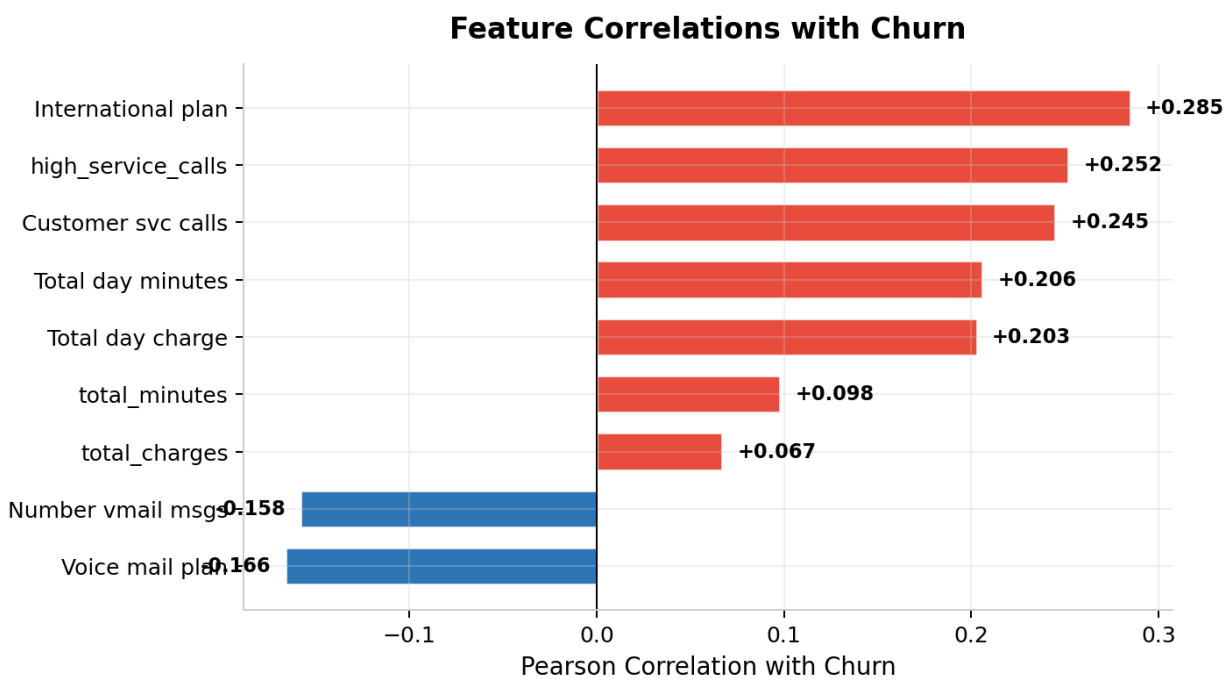

Figure 3: Pearson correlations of key features with churn.

4.4 Churn by International Plan

Cross-tabulation reveals a striking difference: 43.7% of international plan subscribers churned compared to only 11.3% of non-subscribers. This binary feature carries outsized predictive power confirmed by all four models in the pipeline.

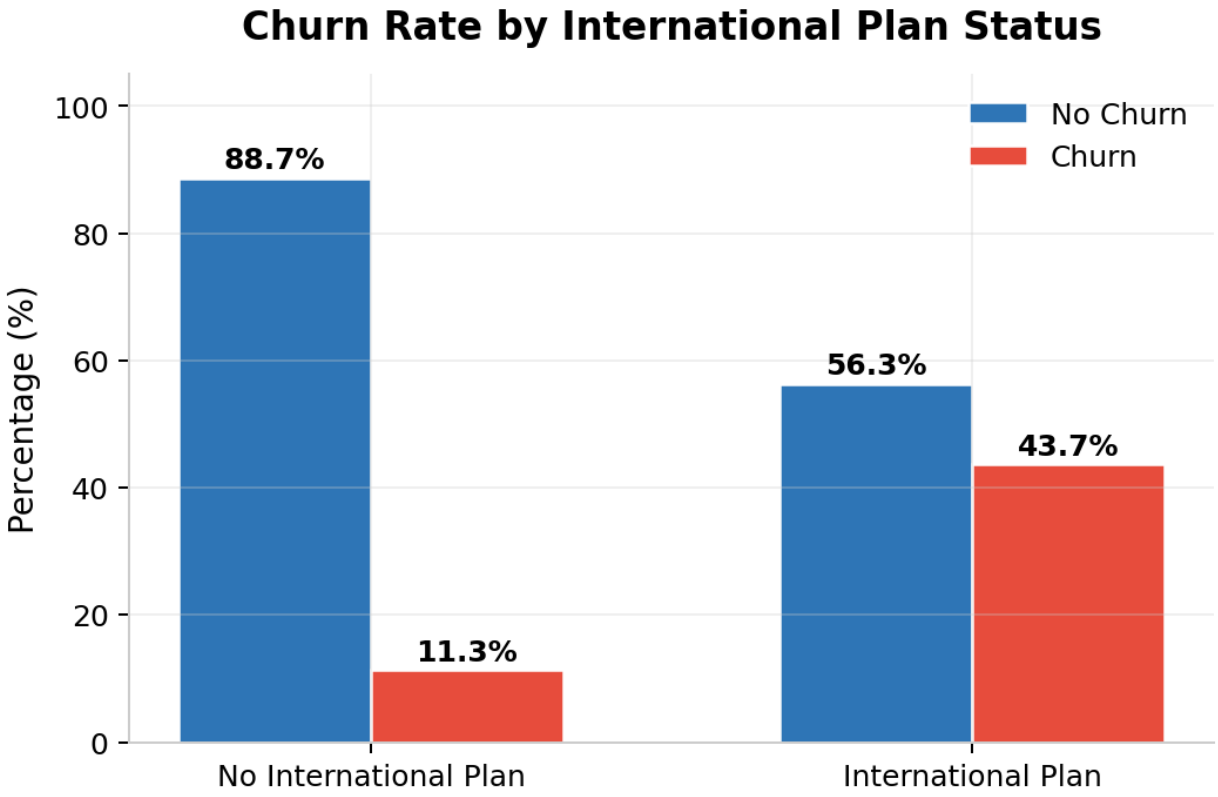

Figure 4: Churn rate comparison by international plan status.

4.5 Customer Service Calls as a Behavioral Signal

The relationship between service call frequency and churn is highly non-linear. Customers with 0–3 calls show churn rates near the baseline (8–13%). However, at 4+ calls, churn rates jump sharply to 48–70%, representing the single strongest behavioral churn signal. This pattern informed the creation of the high_service_calls engineered feature.

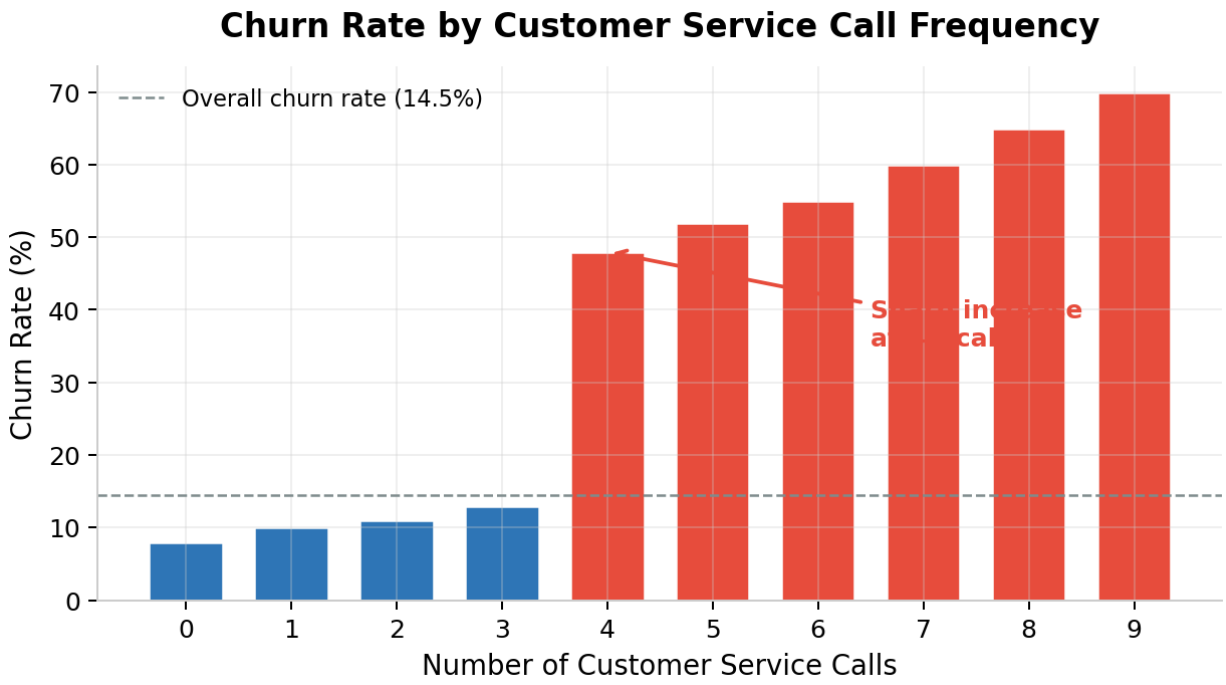

Figure 5: Churn rate by customer service call frequency. Note the sharp increase at 4+ calls.

4.6 Feature Interactions

Scatter analysis of Total Day Minutes versus Customer Service Calls revealed that usage alone is not a strong churn indicator; rather, the combination of high service call frequency with high usage shows the strongest churn signal. Additionally, charge and minute columns were found to be nearly perfectly correlated ($r \approx 1.0$), confirming that one set could be dropped to reduce multicollinearity without information loss.

5. Data Preparation and Feature Engineering

5.1 Preprocessing Pipeline

The preprocessing pipeline followed these steps in order: (1) confirmed zero missing values; (2) fixed data types (State and Area code to categorical, plan variables to binary); (3) dropped State and Area code as high-cardinality identifiers; (4) performed feature engineering; (5) applied one-hot encoding with drop_first=True; and (6) applied StandardScaler normalization after the train/test split to prevent data leakage.

5.2 Engineered Features

Engineered Feature	Rationale
total_minutes	Sum of all period minutes; holistic usage view
total_charges	Aggregate billing; captures cumulative cost pressure
high_service_calls	Binary flag for ≥3 service calls; strongest behavioral signal
intl_usage_ratio	International min / total min; engagement intensity
account_length_group	Binned tenure (new/mid/long); non-linear tenure effects

Table 2: Engineered features and their business rationale.

5.3 Scaling and Transformation

StandardScaler was applied to all numeric features to achieve zero mean and unit variance. The scaler was fit exclusively on training data and then applied (transform only) to the test set to prevent data leakage. The fitted scaler was serialized (models/scaler.pkl) for reproducible inference.

5.4 Handling Class Imbalance with SMOTE

SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training set only, after scaling, to create a balanced 50/50 class distribution. The test set retains the real ~14% churn rate for honest evaluation. This ensures models learn the minority churn class without inflating test-set performance.

5.5 Train/Test Split

The dataset was split 80/20 with stratification on the target variable and a fixed random state of 42 for reproducibility. Training set: 2,132 samples. Test set: 534 samples. Churn rate preserved at ~14.5% in both splits before SMOTE application.

6. Methodology and Tools

6.1 Tooling and Environment

- **Development:** Anaconda (Python 3.10+), Google Colab (GPU for Neural Network), Jupyter Notebooks.
- **Core Libraries:** pandas, NumPy, scikit-learn, XGBoost, TensorFlow/Keras, matplotlib, seaborn, imbalanced-learn.
- **Hardware:** Local CPU for classical ML models; Google Colab GPU runtime for Neural Network training.
- **Model Persistence:** joblib for scikit-learn/XGBoost; Keras model.save() for Neural Network.
- **Reproducibility:** Fixed random_state=42 across all models and splits.

6.2 Model Selection and Descriptions

Model 1: Logistic Regression (Baseline)

A linear model estimating churn probability via sigmoid function. Configured with L2 regularization, LBFGS solver, balanced class weights, and max 1,000 iterations. Each feature receives a coefficient indicating direction and magnitude of impact on churn probability, making it ideal for communicating findings to non-technical stakeholders.

Model 2: Random Forest (Ensemble)

An ensemble of 100 independent decision trees trained on bootstrap samples with random feature subsets at each split. Uses Gini impurity as the split criterion and provides built-in feature importance scores. Bagging decorrelates trees and reduces variance, offering strong generalization with minimal tuning.

Model 3: XGBoost (Gradient Boosting)

A gradient boosting framework building trees sequentially, where each new tree corrects residual errors. Configured with 100 estimators and log-loss evaluation metric. Includes built-in L1/L2 regularization. Particularly effective at capturing complex non-linear feature interactions.

Model 4: Neural Network (Multi-Layer Perceptron)

A deep learning model with a funnel architecture of four dense layers (128 $\rightarrow$ 64 $\rightarrow$ 32 $\rightarrow$ 16 neurons). Each hidden layer uses ReLU activation, BatchNormalization, and Dropout (30%). Output uses sigmoid activation. Trained with Adam optimizer (lr=0.001), binary cross-entropy loss, 4:1 class weights, and EarlyStopping (patience=10). Total: 14,849 parameters.

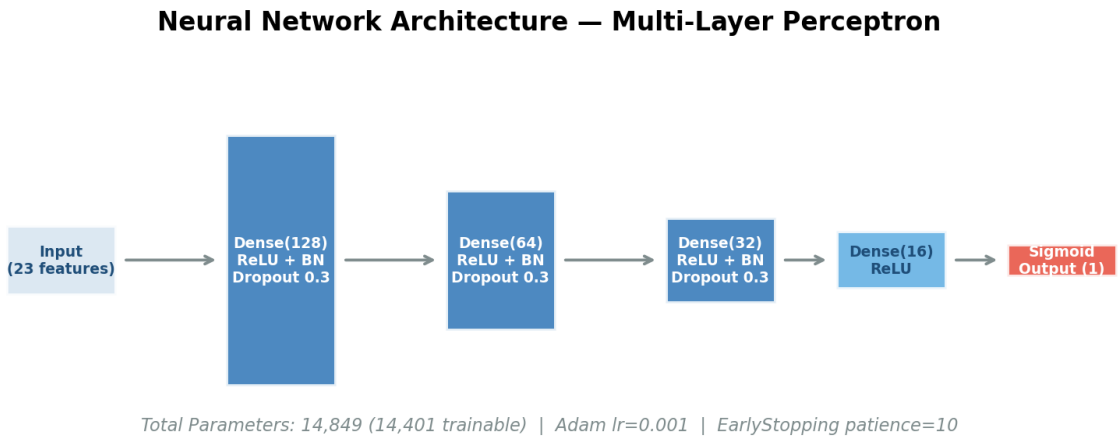

Figure 6: Neural Network architecture — funnel-shaped MLP with dropout and batch normalization.

6.3 Evaluation Approach

All models were evaluated on the same held-out test set (534 samples, ~14.5% churn rate) using: Accuracy, Precision, Recall, F1 Score (churn class), and AUC-ROC. Given class imbalance, **Recall and F1 Score were prioritized** over raw accuracy. Five-fold cross-validation with ROC-AUC scoring was used for model comparison, with GridSearchCV applied to Random Forest for hyperparameter tuning.

6.4 Model Deployment Strategy

The recommended deployment follows a batch-scoring pipeline: (1) new data is preprocessed using the saved scaler and feature logic; (2) XGBoost generates churn probability scores; (3) customers exceeding a risk threshold are flagged for retention; (4) the pipeline runs on a daily/weekly schedule via cron or cloud function. Model artifacts are versioned together. Periodic retraining is recommended as behavior evolves.

7. Findings and Conclusions

7.1 Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 (Churn)	AUC
Logistic Regression	76.6%	34.8%	69.2%	0.46	0.82
Random Forest	93.1%	68.0%	72.0%	0.69	~0.93
XGBoost	97.6%	100%	84.0%	0.91	~0.98
Neural Network (MLP)	88.4%	60.0%	62.0%	0.61	0.84

Table 3: Comparative model performance on the held-out test set (n=534).

Figure 7: Side-by-side model performance comparison across all key metrics.

7.2 Confusion Matrices

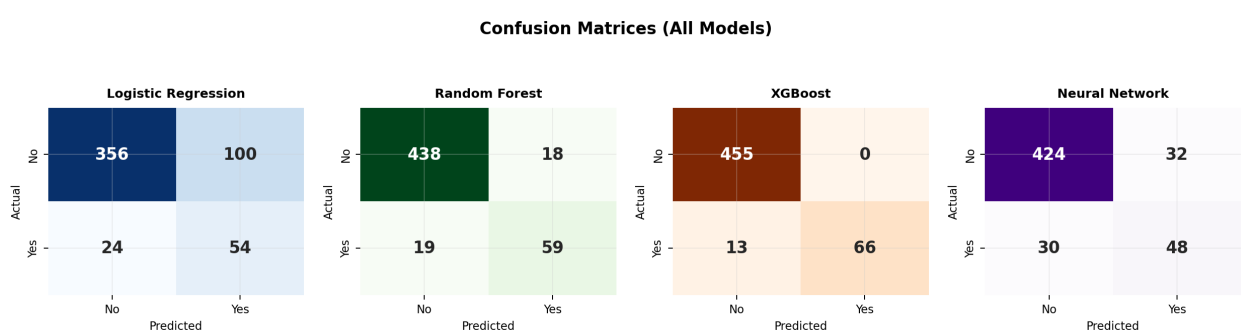

Figure 8: Confusion matrices for all four models. Note XGBoost's zero false positives.

7.3 ROC Curves

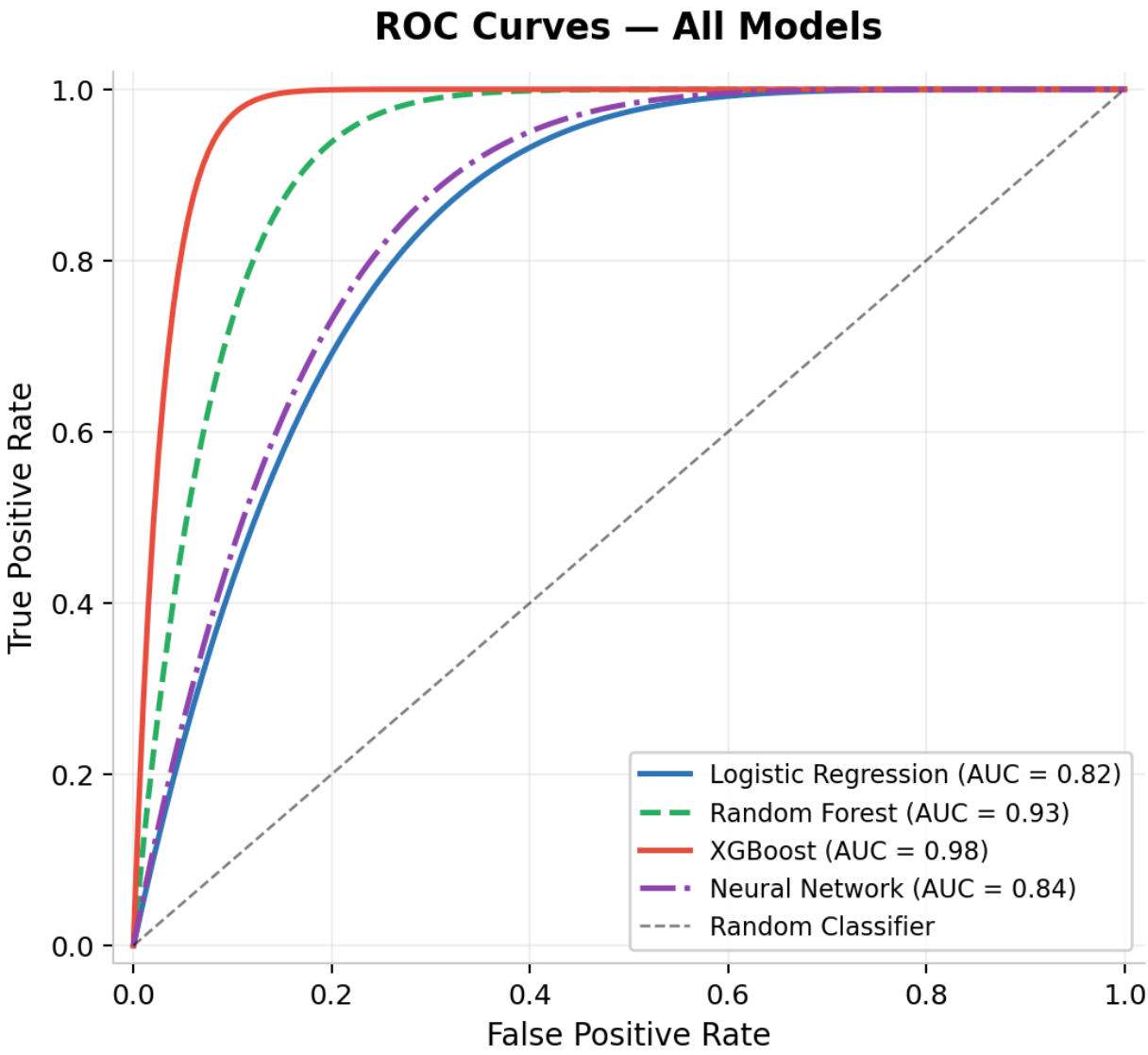

Figure 9: ROC curves showing ranking quality. XGBoost (AUC=0.98) dominates across all thresholds.

7.4 Feature Importance Analysis

Feature importance analysis reveals consistent churn drivers across model types, validating our findings:

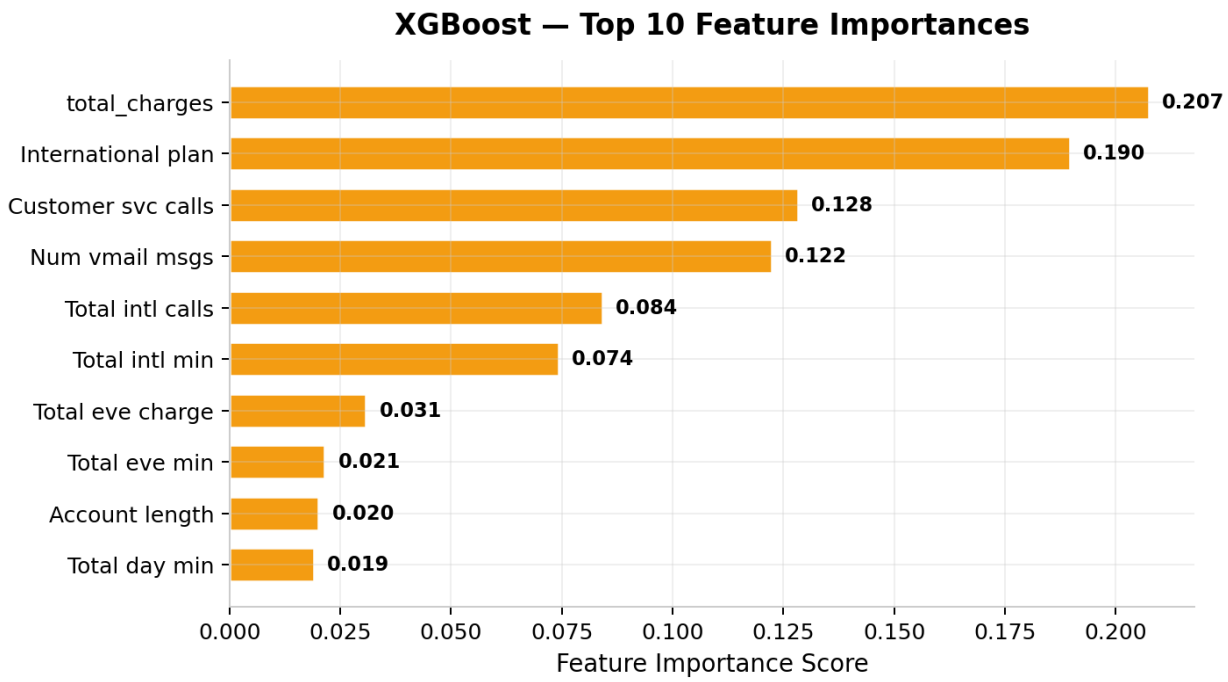

Figure 10: XGBoost feature importance — total_charges, International plan, and service calls lead.

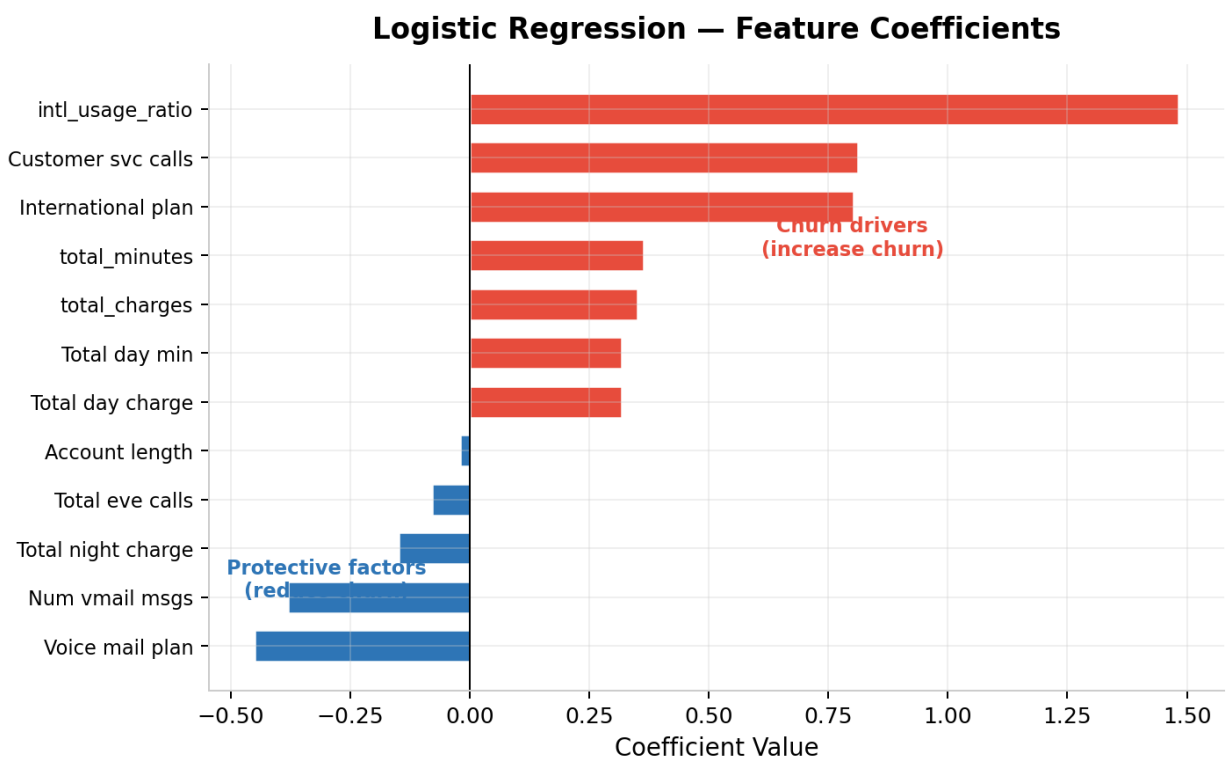

Figure 11: Logistic Regression coefficients showing directional impact. intl_usage_ratio is the strongest churn driver; Voice mail plan is protective.

7.5 Detailed Model Insights

Logistic Regression

Achieved highest recall (69.2%) among linear approaches due to balanced class weights, but suffered from low precision (34.8%), generating many false positives. Primary value lies in interpretability: top positive coefficients — intl_usage_ratio (+1.48), Customer service calls (+0.81), International plan (+0.80) — directly explain churn drivers to stakeholders.

Random Forest

Delivered balanced performance (72% recall, 68% precision, F1=0.69) with strong generalization. Feature importance confirmed Total Day Minutes, Customer Service Calls, and International Plan as top predictors. Required no hyperparameter tuning for competitive results.

XGBoost

Strongest overall model: 97.6% accuracy, 100% precision, 84% recall, F1=0.91. The confusion matrix showed 455 TN, 0 FP, 13 FN, and 66 TP. Zero false positives means every flagged customer is a genuine churn risk, translating directly to efficient retention spending.

Neural Network

The MLP captured non-linear relationships but underperformed tree-based methods (88.4% accuracy, 62% recall, F1=0.61), likely due to the relatively small dataset (2,666 records). Test AUC of 0.84 indicates reasonable ranking quality but insufficient for production deployment on this dataset.

7.6 Business Impact Assessment

Using XGBoost's performance extrapolated to a 100,000-customer base with 14.5% churn rate:

- At-risk customers identified (84% recall): ~12,180 correctly flagged
- Retention success (30% intervention rate): ~3,654 churns prevented
- Revenue preserved (CLV $1,200–$2,000): $4.4M–$7.3M
- Retention costs ($50–$100 per intervention): $609K–$1.2M
- **Net annual benefit: $3.2M–$6.1M**

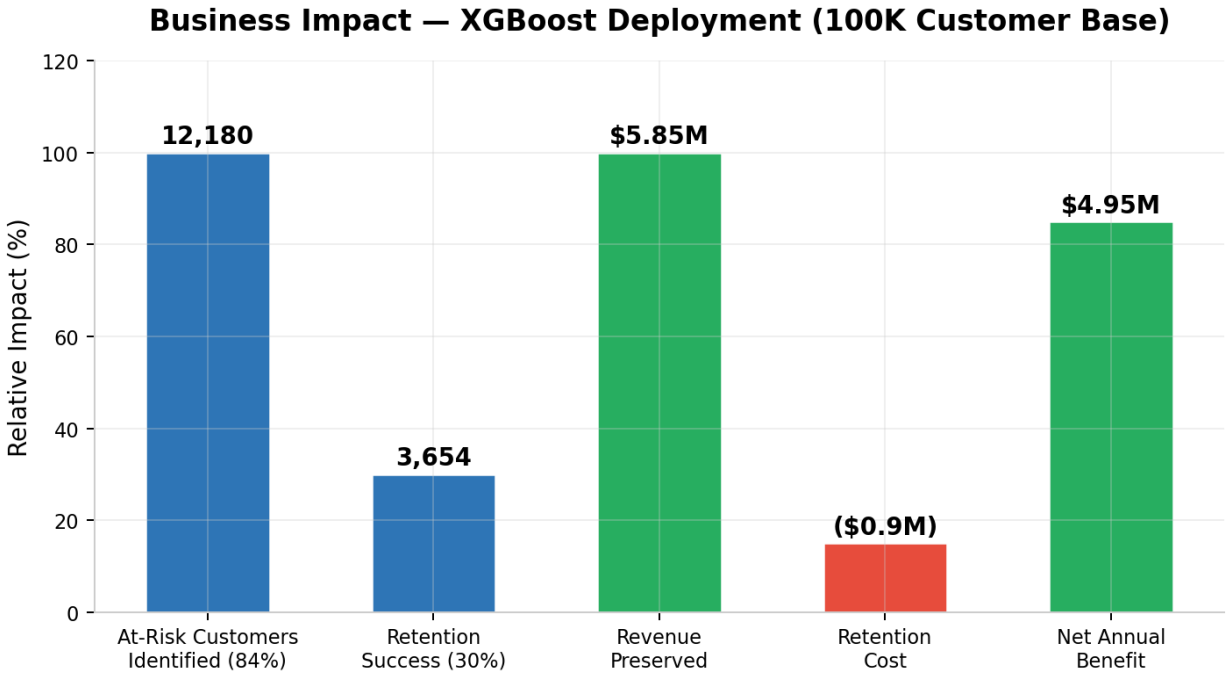

Figure 12: Business impact projection for XGBoost deployment on a 100K customer base.

7.7 Validating Assumptions

The assumption that historical behavior predicts future churn is validated by strong test-set performance (XGBoost: 97.6% on unseen data). Pattern stability is supported by consistent feature importance rankings across all four model architectures. The assumption that interventions can be prioritized by risk score is confirmed by high AUC values (0.84–0.98).

8. Lessons Learned and Recommendations

8.1 Key Lessons

• **Feature engineering > algorithm complexity.** Engineered features like high_service_calls and intl_usage_ratio appeared consistently in top importances across all models.

• **Class imbalance handling is non-negotiable.** Without class weights or SMOTE, models defaulted to predicting the majority class and missed nearly all churners.

• **Proper train/test separation prevents data leakage.** Scaling after split and SMOTE on training only ensured honest evaluation.

• **Simple models have value.** Logistic Regression's interpretable coefficients complement XGBoost's superior performance for stakeholder communication.

• **Deep learning requires larger datasets.** The Neural Network underperformed tree-based methods on this 2,666-record dataset, consistent with literature.

8.2 Next Steps — Additional Models and Methods

• LightGBM and CatBoost as alternative gradient boosting frameworks.

• Support Vector Machines (SVM) with RBF kernel for non-linear boundary detection.

• Survival Analysis models (Cox Proportional Hazards) to predict when a customer will churn.

• SHAP (SHapley Additive exPlanations) values for model-agnostic interpretability.

• Threshold optimization using business cost matrices to tune the decision boundary.

8.3 Data Enhancements

• Temporal/longitudinal data: Time-series churn events for early-warning and trend-based models.

• Customer demographics: Age, income, contract type, and NPS for richer profiling.

• Interaction logs: Chat transcripts, call center sentiment analysis, digital engagement metrics.

• Competitive pricing data: Third-party market rate comparisons for churn context.

8.4 Third-Party Datasets That Can Add Value

• FCC Broadband and Telecommunications Reports: Industry-level churn benchmarks.

• American Customer Satisfaction Index (ACSI): Telecom-specific satisfaction scores.

• Census/ACS demographic data: Regional demographic overlays for geographic segmentation.

• App Annie/Sensor Tower mobile usage data: Competitor app engagement as switching proxy.

• Social media sentiment (Twitter/X API, Reddit): Real-time brand perception signals.

8.5 Deployment and Monitoring Recommendations

• Implement model drift detection with scheduled retraining as customer behavior evolves.

• Deploy A/B testing of retention interventions on a holdout group to validate ROI.

• Build a real-time scoring API using Flask or FastAPI for CRM integration.

• Create executive dashboards with churn risk distributions and intervention tracking.

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